



Pimpri Chinchwad Education Trust's
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Affiliated to Savitribai Phule Pune University(SPPU)

Department of Information Technology

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CipherDocs: A Secure AI-Powered Solution to Prevent Forgery of Critical and Official Documents Using Polygon Blockchain-Based Encryption and Verification

B.Tech Project

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Introduction

With the rapid growth of digital documents across education, government, and enterprise systems, there has been a major shift from traditional paper-based records to digitally issued certificates. This transformation has increased the demand for fast, reliable, and remote document verification. As digital governance and e-services continue to expand, trust and authenticity have become critical factors. At the same time, the widespread use of online document sharing has also increased the risk of document forgery and unauthorized manipulation

Problem Statement

Rising cases of document forgery and unauthorized tampering pose serious challenges to digital systems. Centralized storage creates single points of failure, while manual verification is slow and error-prone. Advanced AI-generated forgeries are difficult to detect, and sensitive user data is often exposed. These issues highlight the need for transparent, tamper-proof, and trustless verification mechanisms.

Motivation

- To eliminate document forgery using decentralized trust mechanisms
- To ensure long-term integrity of issued digital documents
- To reduce dependency on manual and human-driven verification
- To enable instant, reliable third-party verification
- To preserve user privacy during authentication
- To design a scalable and practical real-world solution

Objectives

- To analyze the limitations of centralized document verification systems and address them using a decentralized approach
- To design a verification workflow that maintains document integrity throughout its lifecycle
- To evaluate the effectiveness of AI-based techniques in detecting forged or tampered digital documents
- To develop a privacy-aware verification mechanism suitable for third-party validation
- To evaluate the scalability and cost efficiency of the proposed system in real-world deployment scenarios

Literature Survey

Sr. No.	Paper Name	Core Technology	Specific Limitations Identified	CipherDocs Solution / Novelty
1	Certifier dapp-decentralized and secured certification system using blockchain	Ethereum DApp	High Gas Fees: Ethereum dependency makes it costly. No AI: Purely hash-based verification.	Polygon Network: Reduces fees significantly. AI Integration: Adds semantic forgery detection.
2	DOC-BLOCK: A blockchain based authentication system for digital documents	Ethereum + IPFS	Binary Validation: No explanation for failure. Privacy Leaks: Public IPFS content visibility.	Explainable AI: Provides reasons for rejection. Context-Aware Encryption: Secures off-chain data.
3	A Comparative Study of Ethereum and Polygon for Implementing NFT-Based Certification Systems	Eth vs. Polygon	L2 Security Trade-offs: Notes security variance. Complexity: Cross-chain management.	Native Polygon Deployment: Optimizes specifically for L2 speed/cost, accepting trade-offs for usability.
4	Docschain: Blockchain-based IoT solution for verification of degree documents.	IoT + OCR	OCR Fragility: Fails with poor scans/lighting. Format Limits: B&W only.	Multimodal AI: Robust to image noise/artifacts. Digital-First: Focuses on born-digital assets alongside scans.

Gap Analysis

- Centralized document storage is vulnerable to tampering and data breaches
- Most blockchain-based solutions focus only on immutability, ignoring usability
- Ethereum-based solutions face high transaction costs and scalability issues
- Existing systems are often complex for non-technical users
- Lack of cost-efficient, scalable, and privacy-preserving verification systems

System Architecture

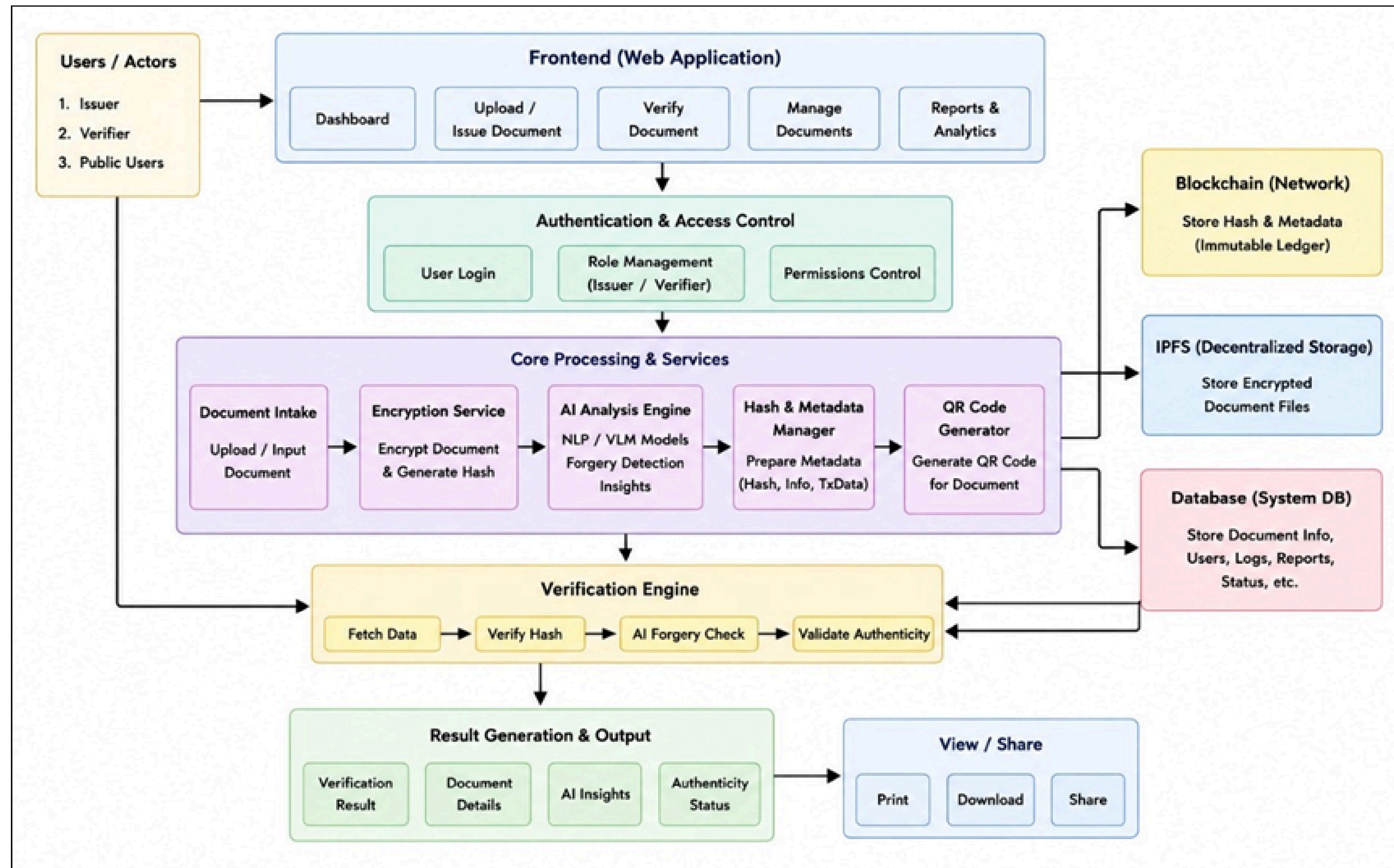


Fig. 1

Research Paper I

- Focuses on the core blockchain architecture of the system
- Uses Polygon Blockchain for fast and low-cost transactions
- Integrates IPFS for secure distributed document storage
- Applies Zero-Knowledge Proofs (ZKP) for privacy-preserving verification
- Implements smart contracts for immutable and transparent record management

Research Paper II

- Focuses on AI as a supportive layer, not the core verification mechanism
- Integrates an intelligent AI chatbot for user assistance and guidance
- Uses logical reasoning models to support decision-making
- Helps in detecting anomalies and inconsistencies in documents
- Improves system usability and interaction through smart automation

From Blockchain Verification to Intelligent Automation

- Paper 1: Secure blockchain-based document authentication
- Limitations: Limited automation and user interaction
- Need for intelligent processing and support
- Introduction of AI-powered extension (Paper 2)
- Integration of agentic AI with blockchain layer

AI-Driven Document Understanding & Processing

- Vision-Language Models for integrity verification
- Automated structured data extraction
- Smart summarization and tagging
- Contextual semantic search
- AI-assisted document export

Performance Monitoring & Secure AI Integration

- AI-driven encryption insights
- Secure processing over blockchain network
- Efficiency and latency monitoring
- System usage analytics dashboard
- Error detection and optimization

AI-Powered Document Understanding & Verification

- Vision-Language Models for document integrity verification
- Automated detection of tampering and inconsistencies
- Structured data extraction from unstructured documents
- Smart summarization for quick understanding
- Context-aware tagging and classification
- Enhances accuracy and reduces manual verification effort

Agentic AI Customer Support & Assistance

- Autonomous AI agents for task management
- AI-powered document chat assistant (Doc Chat)
- Natural language query handling
- Personalized onboarding and guidance
- AI-assisted content generation
- Improves user experience through intelligent interaction

Secure AI Processing & Performance Analytics

- AI-driven encryption and security insights
- Secure processing over blockchain infrastructure
- Real-time monitoring of verification activities
- Efficiency and resource utilization tracking
- Error detection and system optimization
- Ensures security, transparency, and reliability

Sustainable Development Goals (SDGs) Addressed

SDG 4 – Quality Education

- Prevents fake academic certificates and forged credentials
- Enables secure and transparent verification of educational documents

SDG 9 – Industry, Innovation and Infrastructure

- Utilizes blockchain, decentralized storage, and AI-assisted verification
- Promotes scalable and secure digital infrastructure for document management

SDG 16 – Peace, Justice and Strong Institutions

- Prevents document forgery and unauthorized modifications
- Enhances transparency and trust in digital governance systems

Conclusion

- Proposed a decentralized framework to prevent digital document forgery
- Leveraged blockchain technology to ensure tamper-proof verification
- Integrated AI-based validation to enhance forgery detection
- Enabled quick and user-friendly verification using QR codes
- Demonstrated a scalable and cost-efficient solution using the Polygon blockchain

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*Thank
you!*